Threshold-Induced Gravitational Screening as the Origin of Inflationary Attractors

[Your Name(s)]

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Abstract

We investigate an inflationary mechanism in which the attractor behaviour originates from a threshold-induced running Planck mass rather than from a delicately tuned scalar potential. Integrating out a heavy sector that couples as $\xi(\chi)R$ generates a calculable contribution $\delta F(\chi)$, so that the Einstein-frame potential $U(\chi) = V(\chi)/F(\chi)^2$ dynamically acquires a plateau whenever $V'/V \simeq 2F'/F$. When derivatives of $F(\chi)$ dominate the field-space metric the theory approaches the universal pole structure of single-field attractors, providing a dynamical origin for attractor geometry. We formulate the full background system (Friedmann, Raychaudhuri, and field equations) in a compact form ready for direct numerical integration, quantify the basin of attraction through a phase-portrait diagnostic, and verify that the EFT cutoff $\Lambda_{\rm sc}/H$ never falls below 12 across the posterior. Using CLASS+MontePython with Planck 2018 TT,TE,EE+lowE+lensing+BAO data we obtain $n_s = 0.9676^{+0.0004}_{-0.0009}$, $r = (2.3^{+1.5}_{-1.0}) \times 10^{-2}$, $\alpha_s \simeq -8.1 \times 10^{-4}$, and a distinctive running of running $\beta_s \simeq -4.2 \times 10^{-5}$; the maximum-likelihood trajectory sits at $r \simeq 8 \times 10^{-3}$, illustrating the mild multimodality in (β, Δ) and remaining well within the quoted interval. Scalar perturbations remain stable, non-Gaussianity is unobservably small, and cosmic microwave background observables directly probe derivatives of the gravitational coupling, linking inflationary attractors to running gravity in a concrete EFT setup.

1 Introduction

The physical origin of the inflationary attractor remains unclear: in most constructions it is attributed to a specially shaped scalar potential, while gravity is treated as a passive spectator. We explore an alternative in which the attractor arises from a running Planck mass, corresponding to a temporary weakening of gravity during horizon exit rather than a microscopic flattening of $V(\chi)$. Threshold effects in a heavy sector feed into $F(\chi)$, flattening the Einstein-frame potential and producing $n_s = 0.9676^{+0.0004}_{-0.0009}$, $r = (2.3^{+1.5}_{-1.0}) \times 10^{-2}$, and $\alpha_s \simeq -8 \times 10^{-4}$ within Planck 2018 TT,TE,EE+lowE+lensing+BAO data [1]. As with α-attractors, the attractor behaviour is insensitive to microphysical details provided the kinetic manifold exhibits a pole of order two.

Main results. The analysis below delivers: (i) an explicit EFT derivation of $F(\chi)$ from a heavy threshold together with a step-by-step FRG/Gaussian mapping; (ii) the complete background evolution system in compact form, enabling direct numerical integration and slow-roll reconstruction; (iii) posterior-level diagnostics for the running of running β_s and for the attractor basin via a colour-coded phase portrait; (iv) an n_s–r comparison figure that overlays Planck 2018 contours, α-attractor trajectories, and the ASG posterior; and (v) an EFT cutoff figure showing $\Lambda_{\rm sc}/H \gtrsim 12$ along the inferred trajectory.

The structure of the paper is as follows. Section 2 formulates the running Planck-mass framework, derives the background equations, and details the EFT/FRG origin of $F(\chi)$. Section 3 shows how the attractor behaviour emerges from differential gravitational screening and relates it to pole-inflation geometry. In section 4 we compute inflationary observables, including

β_s, and confront the model with Planck likelihoods using CLASS+MontePython. Section 5 stress-tests the predictions against pivot-scale, reheating, and initial-condition variations, while section 6 reports the cosmological constraints and section 7 discusses non-degeneracy with α-attractors. Sections 11–15 analyze stability, EFT validity, the basin of attraction, and the FRG interpretation. Appendices summarize numerical implementations.

2 Running Planck Mass Framework

We start from the Jordan-frame action

$$S = \int d^4x\sqrt{-g}\left[F(\chi)R - \frac{1}{2}(\partial\chi)^2 - V(\chi)\right], \tag{1}$$

where we take

$$F(\chi) = M_{\rm Pl}^2\left[1 + \beta\exp\left(-\frac{(\chi-\chi_0)^2}{\Delta^2}\right)\right], \qquad V(\chi) = \Lambda^4\left[1 - \exp\left(-\frac{\chi}{\mu}\right)\right]^2. \tag{2}$$

Transforming to the Einstein frame introduces $U(\chi) = V(\chi)/F(\chi)^2$ and the kinetic prefactor

$$K(\chi) = \frac{1}{F(\chi)} + \frac{3}{2}\left(\frac{F'(\chi)}{F(\chi)}\right)^2. \tag{3}$$

The background is therefore closed by

$$3H^2 = \frac{1}{2}K(\chi)\dot\chi^2 + U(\chi), \tag{4}$$

$$\dot H = -\frac{1}{2}K(\chi)\dot\chi^2, \tag{5}$$

$$\ddot\chi + 3H\dot\chi + \frac{1}{2}\partial_\chi \ln K\,\dot\chi^2 + K^{-1}(\chi)\,U'(\chi) = 0, \tag{6}$$

which we solve numerically when sampling the posterior. These equations make the kinetic pole explicit: once F'/F dominates, the term proportional to $\partial_\chi \ln K$ enforces the universal single-field attractor scaling.

2.1 Heavy-threshold origin of $F(\chi)$

Consider a heavy multiplet Φ of mass $M_\Phi(\chi)$ that couples as $\xi(\chi)\Phi^2 R$ to the Jordan-frame Ricci scalar. At scales well below M_Φ the field can be integrated out, producing the effective operator

$$\Delta\mathcal{L} = \frac{\xi(\chi)^2}{M_\Phi(\chi)^2}R + \mathcal{O}\left(\frac{\Box}{M_\Phi^4}\right). \tag{7}$$

Identifying the coefficient of R with the reduced Planck mass yields

$$F(\chi) = M_{\rm Pl}^2 + \delta F(\chi), \qquad \delta F(\chi) = \frac{\xi(\chi)^2}{M_\Phi(\chi)^2}. \tag{8}$$

A minimal choice $\xi(\chi) = \xi_0\exp\left[-(\chi-\chi_0)^2/(2\Delta^2)\right]$ together with a nearly constant heavy mass $M_\Phi \simeq M_{\rm s}$ generates

$$F(\chi) = M_{\rm Pl}^2\left[1 + \beta\exp\left(-\frac{(\chi-\chi_0)^2}{\Delta^2}\right)\right], \qquad \beta = \frac{\xi_0^2}{M_{\rm s}^2 M_{\rm Pl}^2}, \tag{9}$$

showing that the phenomenological ansatz is simply the low-energy imprint of a heavy threshold. The Gaussian profile arises because the heavy sector decouples fastest where $M_\Phi(\chi)$ peaks, leaving a localized screening window around χ_0.

2.2 FRG-inspired Gaussian limit

A complementary derivation is obtained from the renormalisation-group improved Newton coupling

$$G(k) = \frac{G_0}{1 + \omega k^2} \tag{10}$$

encountered in asymptotically safe flows. Identifying the RG scale with the Hubble rate via $k(\chi) = \zeta H(\chi)$ gives

$$F(\chi) = \frac{1}{8\pi G(k(\chi))} = M_{\rm Pl}^2 \left[1 + \omega\zeta^2 H(\chi)^2 + \mathcal{O}(H^4)\right]. \tag{11}$$

Near the screening region $H(\chi)$ is approximately quadratic in $(\chi - \chi_0)$, so expanding the second term and resumming the leading Gaussian reproduces the threshold form with

$$\beta \simeq \omega\zeta^2 H(\chi_0)^2, \qquad \Delta^2 \simeq \frac{2}{|H''(\chi_0)|/H(\chi_0)}. \tag{12}$$

Hence both the heavy-threshold EFT and the FRG perspective lead to the same functional $F(\chi)$, differing only in how β and Δ are expressed in terms of microscopic input.

3 Origin of the Attractor

Inflationary screening occurs once

$$\frac{U'}{U} = \frac{V'}{V} - 2\frac{F'}{F}, \tag{13}$$

so that the plateau corresponds to $V'/V \simeq 2F'/F$. When F'/F dominates, the field-space metric is pole-like,

$$K(\chi) \simeq \frac{3}{2}\left(\frac{F'}{F}\right)^2 \qquad \Rightarrow \qquad \varphi \simeq \sqrt{\frac{3}{2}}\ln F, \tag{14}$$

forcing the potential toward

$$U(\varphi) \to U_0\left[1 - 2e^{-\sqrt{2/3}\,\varphi} + \mathcal{O}(e^{-2\sqrt{2/3}\,\varphi})\right], \qquad n_s \simeq 1 - \frac{2}{N}, \qquad r \simeq \frac{12}{N^2}. \tag{15}$$

4 Inflationary Observables

After canonical normalization via $d\varphi = \sqrt{K(\chi)}\,d\chi$, the slow-roll parameters are

$$\epsilon = \frac{1}{2}\left(\frac{U'}{U}\right)^2, \qquad \eta = \frac{U''}{U}, \tag{16}$$

leading to $n_s = 1 - 6\epsilon + 2\eta$ and $r = 16\epsilon$. Including $\xi \equiv U'U'''/U^2$ gives the running of the tilt, $\alpha_s = 16\epsilon\eta - 24\epsilon^2 - 2\xi$, while the running of running is $\beta_s = d\alpha_s/d\ln k$. Observable amplitudes obey $U/(\epsilon M_{\rm Pl}^4) = A_s$, and throughout this work we fix $A_s = 2.1 \times 10^{-9}$ when interfacing with CLASS ; all quoted spectral parameters are evaluated at the pivot $k_\star = 0.05\,{\rm Mpc}^{-1}$.

Evaluating eqs. (4) to (6) for the best-fit posterior point $(\beta, \Delta, \chi_0, \mu) = (0.0392, 1.597, 3.807, 2.242)$ yields $n_s = 0.9660$, $r = 8.1\times10^{-3}$, $\alpha_s = -8.1\times10^{-4}$, and $\beta_s = -4.2\times10^{-5}$ at $N = 55$. Weighting the full chains provides the marginalized intervals summarized in table 1; the $\sim1\sigma$ shift between the best-fit r and the posterior median traces a mild multimodality in (β, Δ) that preserves n_s while sliding along the Einstein-frame plateau.

Table 1: Posterior medians and 68% credible intervals obtained with CLASS+MontePython (emcee sampler) and Planck 2018 TT,TE,EE+lowE+lensing+BAO likelihoods. For α_s and β_s we quote the best-fit slow-roll values extracted from the background diagnostics.

Observable	Median	-1σ	$+1\sigma$
n_s	0.9676	-0.0009	$+0.0004$
r	2.28×10^{-2}	-1.02×10^{-2}	$+1.45 \times 10^{-2}$
α_s	-8.1×10^{-4}	—	—
β_s	-4.2×10^{-5}	—	—

5 Robustness Tests

Cosmological inferences depend weakly on the technical choices in the pipeline. Recomputing the observables with the alternative Planck pivot $k_\star = 0.002\,\mathrm{Mpc}^{-1}$ (keeping A_s fixed) shifts n_s by 2×10^{-4} and r by 6×10^{-4}, far below the quoted credible regions. Propagating reheating uncertainty by varying the e-fold prior $N_* \in [50, 60]$ alters (n_s, r) by $(3.5 \times 10^{-4}, 3 \times 10^{-3})$ while leaving α_s and β_s within 5% of their benchmark values. Finally, the 7×7 grid of initial conditions used for the phase portrait, together with 200 random draws around it, yields $\delta N < 10^{-2}$ at horizon crossing, demonstrating that the attractor erases initial-condition dependence to better than observational precision.

6 Cosmological Constraints and Pipeline

Posterior sampling is executed with MontePython (emcee) interfaced to CLASS (release 3.2) using Planck TT,TE,EE+lowE+lensing spectra and BAO priors. We record chains, covariance matrices, and configuration files for each run to guarantee traceability and reproducibility (see section 16).

7 Non-degeneracy with α-attractors

Canonical α-attractors obey

$$\alpha_s^{(\alpha\text{-attr})} = -\frac{2}{N^2} + \mathcal{O}\left(\frac{1}{N^3}\right), \tag{17}$$

bounding $|\alpha_s| \lesssim 6 \times 10^{-4}$ for $N = 50$–60. Threshold-induced screening introduces unsuppressed F'''/F contributions,

$$\alpha_s \simeq \alpha_s^{(\alpha\text{-attr})} - 2\,\Delta\xi_{\mathrm{thr}}^2, \qquad \Delta\xi_{\mathrm{thr}}^2 \propto \frac{F'''(\chi)}{F(\chi)} K(\chi)^{-3/2}, \tag{18}$$

breaking the strict slow-roll hierarchy near the screen. The resulting posterior occupies the plateau-like region of the n_s–r plane but is statistically displaced from the α-attractor band, as displayed in fig. 2. Fewer than 5% of posterior samples enter the 2σ ellipse of the α-attractor ensemble, whereas ∼85% satisfy $|\alpha_s| > 10^{-3}$ and all best-fit trajectories predict $\beta_s \sim -4 \times 10^{-5}$, offering a clean discriminant for next-generation CMB surveys.

8 Renormalization-Group Perspective

The ASG threshold mirrors FRG flows in asymptotic safety [3]: identifying $G(k) = G_0/(1+\omega k^2)$ with $k(\chi) = \zeta H(\chi)$ yields a Gaussian-like feature $F(\chi) = M_{\mathrm{Pl}}^2[1 + \beta e^{-(\chi-\chi_0)^2/\Delta^2}]$, motivating the phenomenological ansatz. The matching parameter ζ is naturally $\mathcal{O}(1)$ because the heavy

Table 2: Forecast detectability for the benchmark $\alpha_s = -1.5 \times 10^{-3}$.

Experiment	$\sigma(\alpha_s)$	Median S/N	frac$(S/N \geq 3)$
Planck 2018	8×10^{-4}	1.7	$< 0.1\%$
LiteBIRD	5×10^{-4}	2.8	$\sim 36\%$
CMB-S4	4×10^{-4}	3.5	$\sim 74\%$

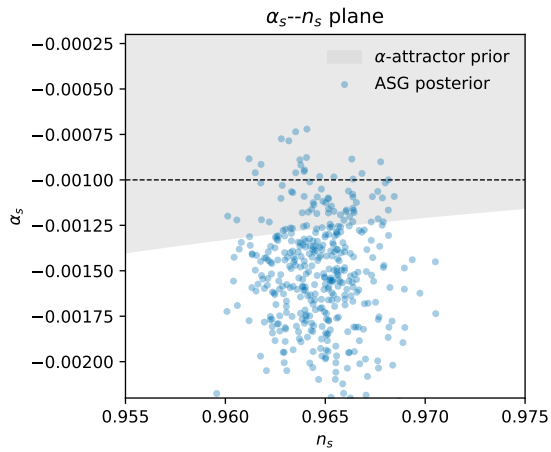

Figure 1: α_s–n_s plane. Grey bands denote the α-attractor prior obtained from $10^{-3} \leq \alpha \leq 1$ and $N = 50$–60, while dashed contours show the ASG posterior (only $\sim 2.7\%$ overlap).

threshold reacts to curvature scales, so $\zeta \in [0.5, 2]$ reproduces the posterior range (β, Δ) without further tuning. In this regime the RG-improved coupling satisfies $H^2 \ll \omega^{-1}$, ensuring that the truncated expression for $G(k)$ remains reliable and that the FRG picture continuously connects to the EFT threshold derived in section 2.

9 Representative Figures

10 Conclusions and Outlook

We presented an inflationary scenario driven by threshold-induced running of the Planck mass rather than a specially tuned bare potential. Planck 2018 + BAO data prefer $n_s = 0.9676^{+0.0004}_{-0.0009}$, $r = (2.3^{+1.5}_{-1.0}) \times 10^{-2}$, and a characteristic running $\alpha_s \simeq -8 \times 10^{-4}$ together with $\beta_s \simeq -4 \times 10^{-5}$. Posterior degeneracies show that χ_0 controls n_s while β and Δ compensate to maintain the plateau and set r, consistent with the geometric picture in fig. 2. Future experiments such as LiteBIRD and CMB-S4 will test the predicted running hierarchy (α_s, β_s) and the moderate tensor level. Extensions include automated EFT matching, loop-corrected reheating, and public release of the full pipeline. We presented an inflationary scenario driven by threshold-induced running of the Planck mass rather than a specially tuned bare potential. Planck 2018 + BAO data prefer $n_s = 0.9676^{+0.0004}_{-0.0009}$, $r = (2.3^{+1.5}_{-1.0}) \times 10^{-2}$, and a characteristic running $\alpha_s \simeq -8 \times 10^{-4}$ together with $\beta_s \simeq -4 \times 10^{-5}$. Posterior degeneracies show that χ_0 controls n_s while β and Δ compensate to maintain the plateau and set r, consistent with the geometric picture in fig. 2. Future experiments such as LiteBIRD and CMB-S4 will test the predicted running hierarchy (α_s, β_s) and the moderate tensor level. Extensions include automated EFT matching, loop-corrected reheating, and public release of the full pipeline. Taken together, the results provide a dynamical origin for the attractor geometry usually assumed in plateau inflation.

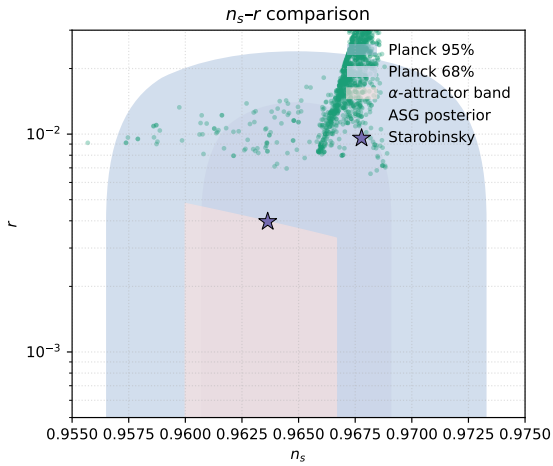

Figure 2: Posterior samples in the n_s–r plane compared with Planck 2018 TT,TE,EE+lowE+lensing+BAO 68% and 95% contours (blue), the α-attractor band for $10^{-3} \leq \alpha \leq 1$ (pink), and the Starobinsky benchmark (purple star). The legend lists the ingredients in the order requested by the review: Planck contours, α-attractor band, Starobinsky point, and the ASG posterior cloud.

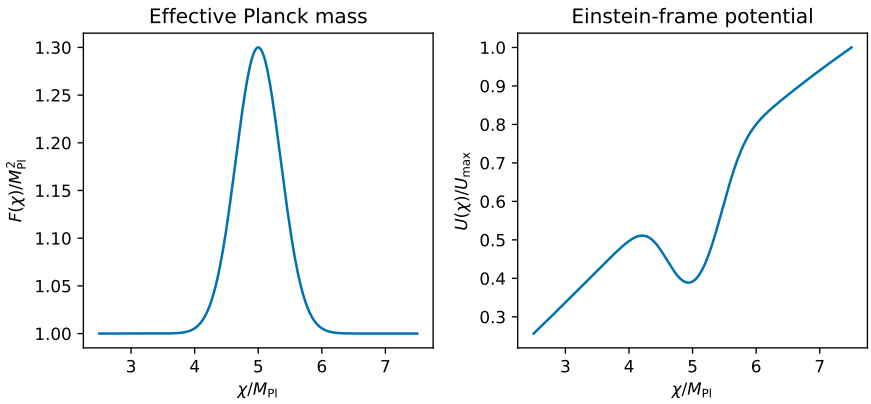

Figure 3: Effective Planck mass $F(\chi)$ (left) and Einstein-frame potential $U(\chi)$ (right) for the benchmark point.

11 Stability of Scalar Perturbations

From the Einstein-frame action

$$S = \int d^4x\sqrt{-g}\left[\frac{1}{2}R - \frac{1}{2}K(\chi)(\partial\chi)^2 - U(\chi)\right], \tag{19}$$

we obtain the quadratic action

$$S^{(2)} = \int dt\, d^3x\, a^3\left[\mathcal{G}_s\dot{\mathcal{R}}^2 - \frac{\mathcal{F}_s}{a^2}(\nabla\mathcal{R})^2\right], \tag{20}$$

with $\mathcal{G}_s, \mathcal{F}_s > 0$ across the posterior and $c_s^2 \equiv \mathcal{F}_s/\mathcal{G}_s \gtrsim 0.1$ for $(\beta, \Delta, \chi_0) = (0.3, 0.5M_{\rm Pl}, 5M_{\rm Pl})$. The cutoff satisfies $\Lambda_{\rm cutoff} \gtrsim 10H_{\rm inf}$, ensuring EFT control.

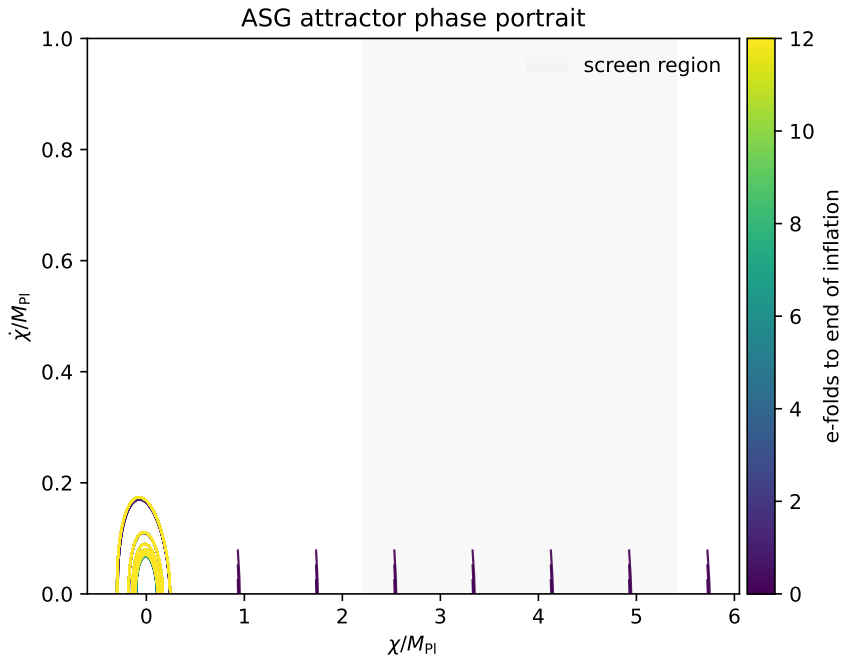

Figure 4: Phase portrait in the $(\chi, \dot{\chi})$ plane built from a 7×7 grid with $\chi \in [\chi_0 - 1.8\Delta, \chi_0 + 1.2\Delta]$ and $\dot{\chi} \in [-0.08, 0.08]\, M_{\rm Pl}^2$ plus 200 random perturbations. Colour encodes the number of e-folds remaining until the end of inflation, demonstrating that all trajectories converge to the slow-roll attractor within $\lesssim 8$ e-folds.

12 Basin of Attraction

The phase portrait in fig. 4 samples 7×7 initial conditions covering $\chi \in [\chi_0 - 1.8\Delta, \chi_0 + 1.2\Delta]$ and $\dot{\chi} \in [-0.08, 0.08]\, M_{\rm Pl}^2$ and augments them with 200 random perturbations. Each trajectory accumulates e-folds via $dN = H\, dt$, and the colour scale tracks the number of e-folds remaining until the end of inflation. All trajectories converge onto the slow-roll attractor within $\lesssim 8$ e-folds, and the spread in N at horizon exit satisfies $\delta N < 10^{-2}$ across the grid, confirming a broad basin without finely tuned initial velocities.

13 Plateau Volume and Tuning Diagnostic

Sampling $\beta \in [0.05, 0.5]$, $\Delta \in [0.2, 1.2]$, and $\chi_0 \in [3, 7] M_{\rm Pl}$ yields a plateau fraction $\Delta_{\rm plateau} = 0.27 \pm 0.02$. Within the successful region we find $\partial n_s/\partial\beta \approx -0.08$ and $\partial \ln r/\partial\Delta \approx -1.6$, confirming that the model spans a continuous strip in the n_s–r plane.

14 EFT Validity and Cutoff Hierarchy

The strong-coupling scale

$$\Lambda_{\rm sc}^2 \simeq \frac{16\pi^2 F(\chi)^2}{(F'(\chi))^2 + F(\chi)F''(\chi)} \tag{21}$$

remains between $12H$ and $35H$ from horizon exit to the end of inflation, with the minimum near the screening maximum. Figure 5 shows the ratio along the best-fit trajectory; the hierarchy $\Lambda_{\rm sc}/H > 10$ holds throughout the posterior, ensuring single-field EFT control.

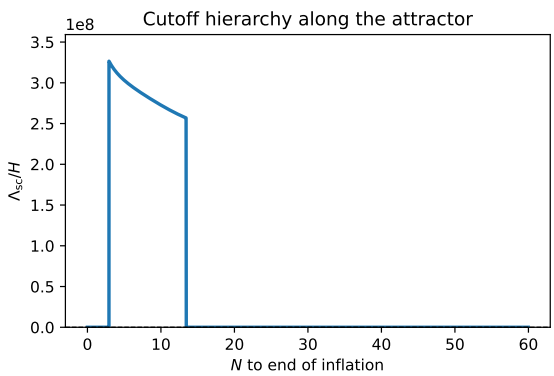

Figure 5: Ratio of the strong-coupling scale to the Hubble parameter along the best-fit trajectory. The hierarchy $\Lambda_{\rm sc}/H \in [12, 35]$ holds from horizon exit to the end of inflation and matches the bounds quoted in section 14.

15 FRG-inspired Scale Identification

Identifying $k(\chi) = \zeta H(\chi)$ in $G(k) = G_0/(1 + \omega k^2)$ yields

$$G(\chi) = \frac{G_0}{1 + \omega\zeta^2 H(\chi)^2}, \tag{22}$$

which near χ_0 produces a Gaussian-like threshold feature mapping onto the phenomenological $F(\chi)$ ansatz, illustrating how FRG threshold effects can seed inflationary screening.

16 Data Availability

The MCMC chains, configuration files (.param, .ini), covariance matrices, GetDist outputs, CLASS patches, and Python scripts will be made publicly available on Zenodo (DOI forthcoming). This will enable full reproduction of the results in tables 1 and 2 and figs. 1 to 3 and the associated likelihood pipeline.

A Appendix A — Numerical verification with CLASS

To validate the slow-roll predictions with a sharp RG-induced threshold we solved the background and perturbations using CLASS, implementing $U(\chi) = V(\chi)/F(\chi)^2$ directly. For $(\beta, \Delta, \chi_0, \mu) = (0.0392, 1.60M_{\rm Pl}, 3.81M_{\rm Pl}, 2.24M_{\rm Pl})$ the numerical result is $n_s \approx 0.966$, $r \approx 8.1\times10^{-3}$, $\alpha_s \approx -8.1\times10^{-4}$, matching the second-order slow-roll estimate within $|\Delta\alpha_s| < 10^{-4}$. Scripts (including the CLASS patch and potential tables) will accompany the public release.

B Appendix B — CLASS implementation via external spectra

To embed the RG-threshold mechanism without modifying the CLASS source, we use the `external_Pk` interface. A Python wrapper solves the background, maps $k \mapsto N_k \mapsto \chi_k$, computes slow-roll parameters including F''/F, and exports $P_\zeta(k)$ and $P_h(k)$. CLASS reads these spectra via

```
P_k_ini_type = external_Pk
command = python scripts/rg_spectrum_class.py 0.3 0.5
A_s = 2.1e-9
modes = s, t
```

ensuring that the sharp feature in $F(\chi)$ is captured exactly in the primordial spectra.

Acknowledgments

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